

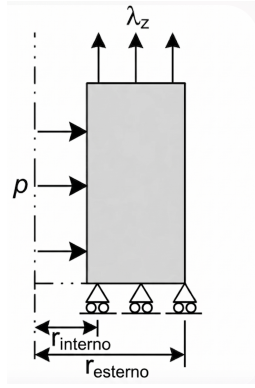


Analysis of a Hyperelastic Cylinder Subject to Internal Pressure and Imposed Axial Deformation

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1 Analytical Model

The problem considered is axisymmetric: the action of internal pressure p and imposed axial elongation λ_z is independent of the coordinate θ , therefore only a cross-section of the cylinder needs to be considered. Let us define $v(r, \theta)$ and $u(r, \theta)$ as the radial and axial displacements of cylinder points, respectively. The preceding considerations allow us to define the following boundary conditions:



$$\begin{cases} v(r, 0) = 0 \\ v(r, H) = \lambda_z H \\ \sigma_{rr}(R_i, z) = -p \\ \sigma_{rr}(R_e, z) = 0 \end{cases}$$

We assume the following transformation:

$$\begin{cases} r = f(R) \\ \theta = \Theta \\ z = \lambda_z Z \end{cases} \rightarrow \vec{x} = f(R)E_r + \lambda_z Z E_z$$

Computing its deformation gradient in cylindrical coordinates:

$$\begin{aligned} A = \nabla \otimes \vec{x} &= \left(E_r \frac{\partial}{\partial R} + E_\theta \frac{1}{R} \frac{\partial}{\partial \Theta} + E_z \frac{\partial}{\partial Z} \right) \otimes (f(R)E_r + \lambda_z Z E_z) \\ &= E_r \otimes E_r f'(R) + E_\theta \otimes E_\theta \frac{f(R)}{R} + E_z \otimes E_z \lambda_z \\ &= \begin{bmatrix} f'(R) & 0 & 0 \\ 0 & \frac{f(R)}{R} & 0 \\ 0 & 0 & \lambda_z \end{bmatrix} \end{aligned}$$

Since in this case $R = I$, the three diagonal terms correspond respectively to radial, circumferential, and axial stretches $\lambda_r, \lambda_\theta, \lambda_z$. Imposing volume conservation:

$$\det(A) = 1 \rightarrow f'(R) \frac{f(R)}{R} \lambda_z = 1$$

Observing that $f(R) = r$ and that $f'(R) = \frac{dr}{dR}$:

$$\begin{aligned} \frac{dr}{dR} \frac{r}{R} \lambda_z = 1 &\rightarrow r dr = \frac{1}{\lambda_z} R dR \\ r &= \sqrt{\frac{R^2}{\lambda_z} + C} \end{aligned}$$

In deriving the right Cauchy-Green tensor, it is convenient to exploit the incompressibility relation to use only two of the three stretches:

$$C = A^T A = \text{diag}(\lambda_r^2, \lambda_\theta^2, \lambda_z^2) = \text{diag}\left(\frac{1}{\lambda_z^2 \lambda_\theta^2}, \lambda_\theta^2, \lambda_z^2\right)$$

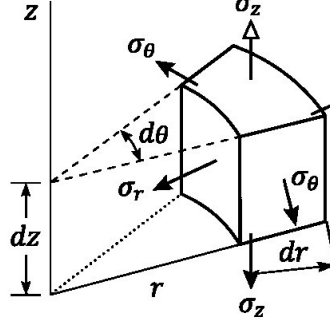
A material is called hyperelastic if a strain energy function W can be defined that, when differentiated, allows us to obtain the stress-strain constitutive equation. For a two-parameter Mooney-Rivlin material, this can be written as follows:

$$\begin{aligned} W(I_1, I_2) &= c_{10}(I_1 - 3) + c_{01}(I_2 - 3) \\ I_1 &= \lambda_r^2 + \lambda_\theta^2 + \lambda_z^2 \\ I_2 &= \lambda_r^2 \lambda_\theta^2 + \lambda_r^2 \lambda_z^2 + \lambda_\theta^2 \lambda_z^2 \end{aligned}$$

where I_1, I_2 are invariants of tensor C . The constitutive relation for the Cauchy stress tensor is as follows (by cylindrical symmetry the left Cauchy-Green tensor equals the right one, $C = A^T A = A A^T = B$):

$$\begin{aligned} \boldsymbol{\sigma} &= -pI + 2 \frac{\partial W}{\partial I_1} B - 2 \frac{\partial W}{\partial I_2} B^{-1} \\ &= -pI + 2c_{10}C - 2c_{01}C^{-1} \\ &= \begin{cases} \sigma_r(R) = -p(R) + 2c_{10} \frac{1}{(\lambda_z \lambda_\theta)^2} - 2c_{01}(\lambda_z \lambda_\theta)^2 \\ \sigma_\theta(R) = -p(R) + 2c_{10} \lambda_\theta^2 - 2c_{01} \frac{1}{\lambda_\theta^2} \\ \sigma_z(R) = -p(R) + 2c_{10} \lambda_z^2 - 2c_{01} \frac{1}{\lambda_z^2} \end{cases} \end{aligned}$$

Considering an infinitesimal element of the axisymmetric cylinder, the equilibrium condition in the radial direction can be derived as follows:



$$\left(\frac{\partial \sigma_r}{\partial r} dr + \sigma_r \right) (r + dr) d\theta dz - \sigma_r r d\theta dz - 2\sigma_\theta \frac{d\theta}{2} dr dz = 0$$

$$\downarrow$$

$$\frac{\partial \sigma_r}{\partial r} + \frac{\sigma_r - \sigma_\theta}{r} = 0$$

Developing the calculations using what was previously derived from the constitutive equation:

$$d\sigma_r = \left(2c_{10} \left(\lambda_\theta^2 - \frac{1}{(\lambda_z \lambda_\theta)^2} \right) + 2c_{01} \left((\lambda_\theta \lambda_z)^2 - \frac{1}{\lambda_\theta^2} \right) \right) \frac{dr}{r}$$

Since $\lambda_\theta = \frac{f(R)}{R}$ it is convenient to express everything as a function of only one of the two parameters. Using λ_θ as the differential, we obtain exclusively polynomial or logarithmic integrals:

$$\frac{dr}{r} = \frac{1}{1 - \lambda_z \lambda_\theta^2} \frac{1}{\lambda_\theta} d\lambda_\theta$$

$$d\sigma_r = \left(2c_{10} \left(\frac{1}{\lambda_z \lambda_\theta} + \frac{1}{\lambda_z^2 \lambda_\theta^3} \right) + 2c_{01} \left(\frac{1}{\lambda_\theta} + \frac{1}{\lambda_z \lambda_\theta^3} \right) \right) d\lambda_\theta$$

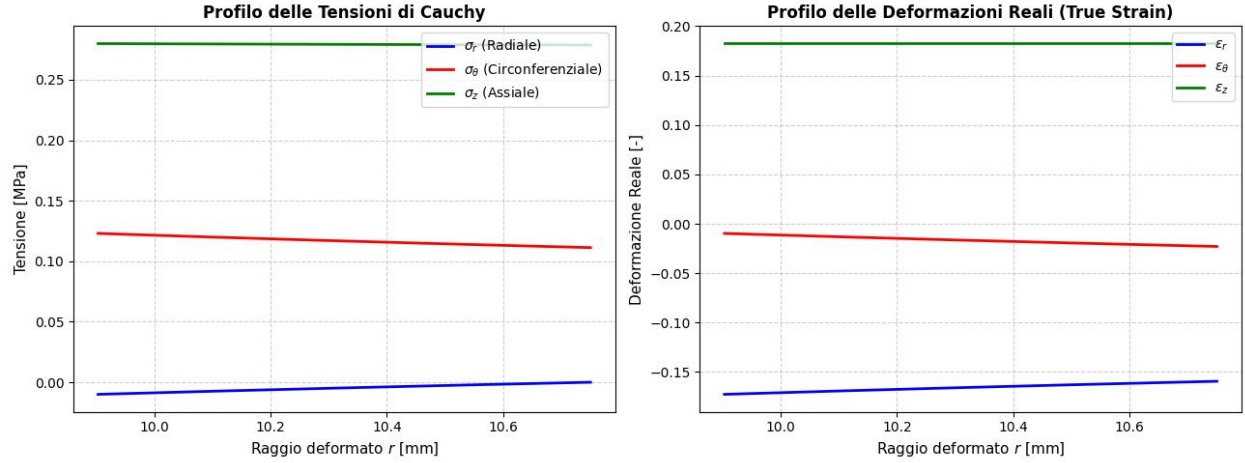
Integrating yields:

$$\sigma_r(\lambda_r) = 2 \left(\frac{c_{10}}{\lambda_z} + c_{01} \right) \ln \lambda_\theta - \left(\frac{c_{10}}{\lambda_z^2} + \frac{c_{01}}{\lambda_z} \right) \frac{1}{\lambda_\theta^2} + c$$

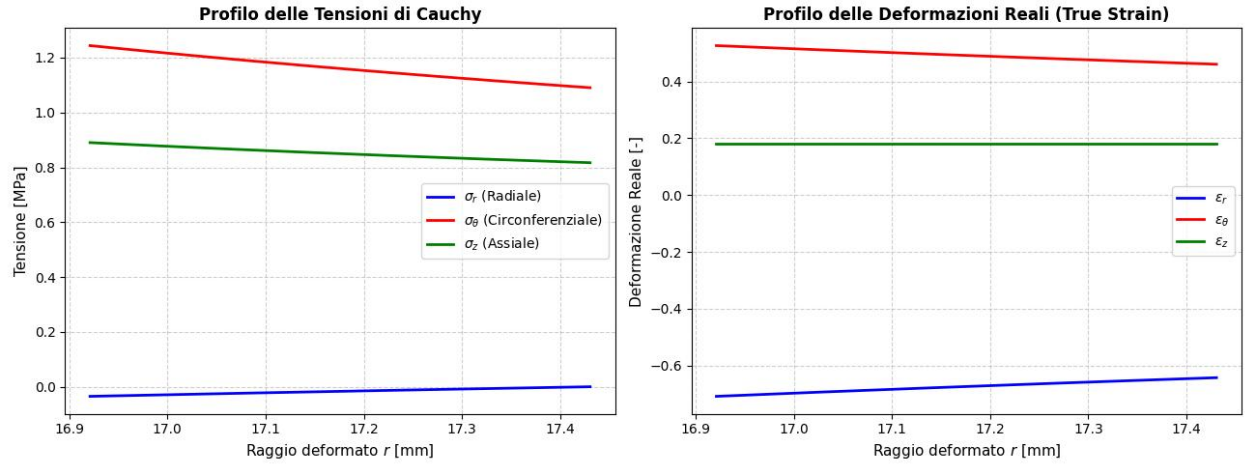
The problem is closed but cannot be solved without numerical methods: imposing the two boundary conditions on stress (internal pressure and zero external stress) yields a transcendental equation:

$$P_i = \left(\frac{C_{10}}{\lambda_z} + C_{01} \right) \left[\ln \left(\frac{R_i^2 + C\lambda_z}{R_e^2 + C\lambda_z} \cdot \frac{R_e^2}{R_i^2} \right) + \frac{C\lambda_z (R_e^2 - R_i^2)}{(R_i^2 + C\lambda_z)(R_e^2 + C\lambda_z)} \right]$$

Imposing an internal pressure of $p_i = 0.01MPa$ and an axial stretch of $\lambda_z = 1.2$ on a cylinder with inner radius $r_i = 10mm$ and outer radius $r_e = 11mm$, the following stress and strain values are obtained (with constants $c_{10} = c_{01} = 0.1MPa$):



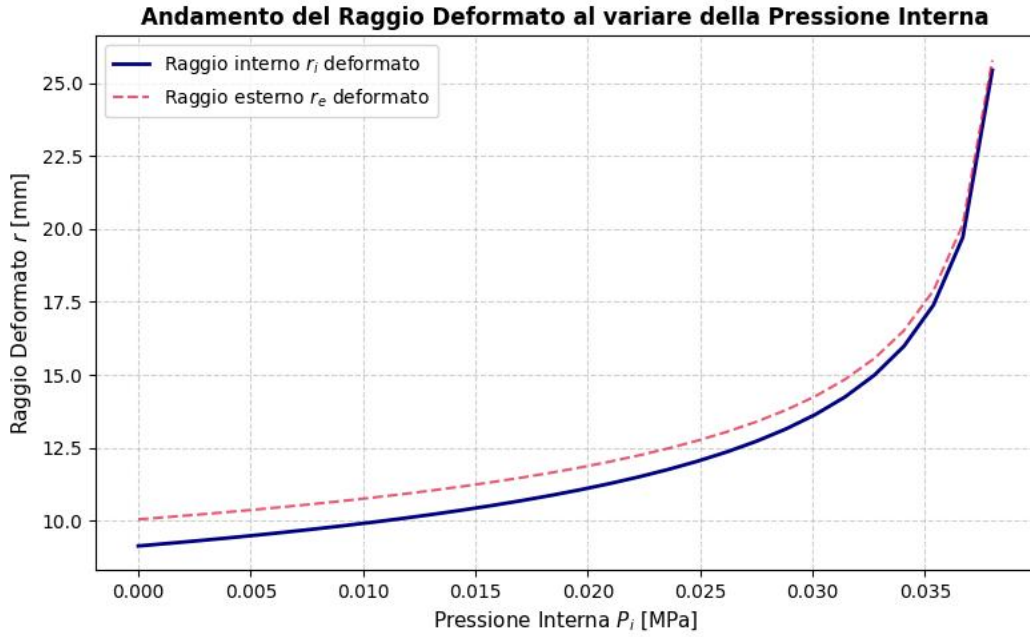
The combination of pressure and imposed stretch in this case is such that the cylinder diameter decreases (negative circumferential strain is observed, despite having tensile hoop stresses) while elongating in the axial direction. Using a higher pressure of $p_i = 0.035MPa$, an increase is observed in both height and diameter, with thickness reduction to satisfy the incompressibility constraint:



Analytically, the limiting value of pressure p_{max} that leads to solution instability, with infinite radial deformation and resulting problem divergence, can be derived. It is:

$$p_{max} = \frac{2(c_{10} + c_{01}\lambda_z^2)}{\lambda_z} \ln\left(\frac{R_e}{R_i}\right) = 0.03876 \text{ MPa}$$

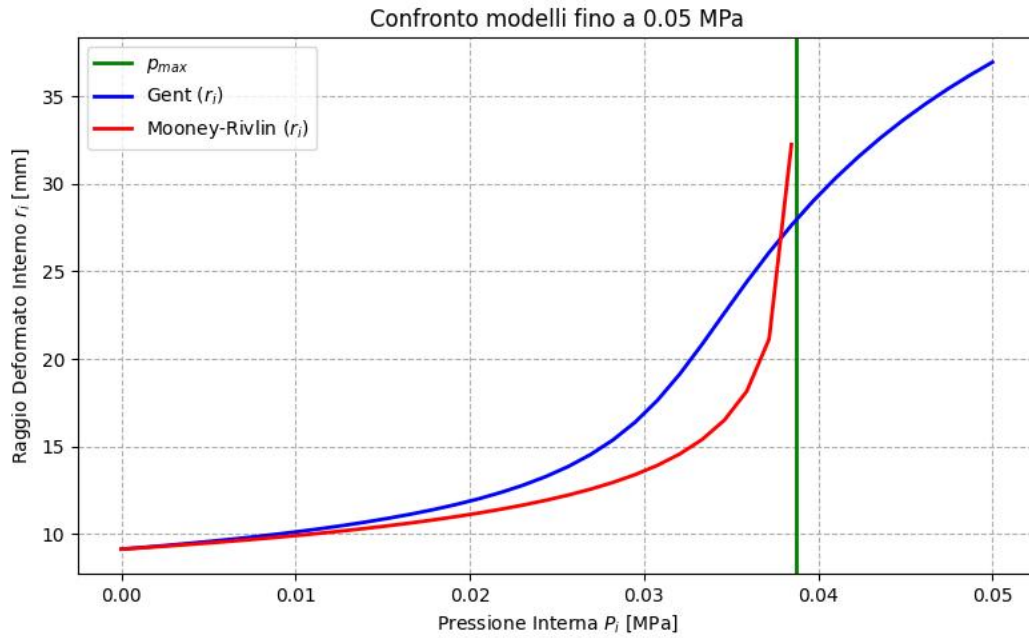
This phenomenon can be observed in the following graph and is intrinsically linked to the constitutive model used, which does not provide material stiffening following high deformations and causes the allowable stress to saturate at a maximum value.



The Gent model is a hyperelastic model that considers material stiffening due to the approach of polymer chains in the large deformation regime. The energy is modeled with an additional parameter I_m that imposes a limit on allowable deformation, while the parameter μ coincides with the shear modulus and is equivalent to $\mu = 2(c_{10} + c_{01})$.

$$W = -\frac{\mu(I_m - 3)}{2} \ln\left(1 - \frac{I_1 - 3}{I_m - 3}\right)$$

Setting $\mu = 0.4 \text{ MPa}$ and $I_m = 30$ (value available in literature for conventional rubbers) yields the following graph:



2 Weak Formulation for Hyperelastic Gent Material (Axisymmetric)

To set up the finite element problem, it is necessary to rigorously derive the virtual work equation. Let us consider the reference domain Ω_0 described by cylindrical coordinates (R, Θ, Z) .

The boundary conditions of the problem are the same as in the analytical model with the sole exception of a notation change: we call p_{in} the pressure applied to the inner face of the cylinder, to distinguish it from p which becomes a Lagrange multiplier to enforce the incompressibility condition.

Gent Constitutive Model

The hyperelastic behavior of the material is governed by the Gent strain energy density function, which models the progressive stiffening due to the strain limit of polymer chains. The constitutive equation is explicitly defined as:

$$W = -\frac{\mu J_m}{2} \ln \left(1 - \frac{I_1 - 3}{J_m} \right) \quad (1)$$

where μ represents the shear modulus at small strains, J_m is the dimensionless parameter that defines the strain limit of the molecular network, and $I_1 = \text{tr}(\mathbf{C}) = \text{tr}(\mathbf{A}^T \mathbf{A})$ is the first invariant of the right Cauchy-Green tensor, with $\mathbf{A}$ being the deformation gradient tensor.

The first Piola-Kirchhoff stress tensor $\mathbf{P}$ is decomposed into a deviatoric part and a volumetric part, within which p acts as a Lagrange multiplier associated with the incompressibility constraint, namely the hydrostatic pressure. Differentiating Gent energy with respect to the deformation gradient $\mathbf{A}$, we obtain the deviatoric part of stress:

$$\mathbf{P}_{dev} = \frac{\partial W}{\partial \mathbf{A}} = \frac{\mu}{1 - \frac{I_1 - 3}{J_m}} \mathbf{A} \quad (2)$$

Consequently, the total stress tensor becomes:

$$\mathbf{P} = \mathbf{P}_{dev} - pJ\mathbf{A}^{-T} = \frac{\mu}{1 - \frac{I_1 - 3}{J_m}} \mathbf{A} - pJ\mathbf{A}^{-T} \quad (3)$$

where $J = \det(\mathbf{A})$.

Derivation of the Weak Form

Introducing test functions, or kinematically admissible variations, $\delta \mathbf{u} = [v_R, v_Z]^T$ for the displacement field and q for the pressure field, the principle of virtual work imposes that the internal virtual work is in perfect equilibrium with the external virtual work. Therefore, the residual equation $\mathcal{R}$ integrated over the reference volume V_0 takes the following form:

$$\int_{V_0} (\mathbf{P}_{dev} \otimes \nabla_\chi \delta \mathbf{u} - p J \mathbf{A}^{-T} \otimes \nabla_\chi \delta \mathbf{u}) dV_0 + \int_{V_0} (J - 1) q dV_0 - \int_{\Gamma_{in}} \mathbf{t}_{ref} \cdot \delta \mathbf{u} dA_0 = 0 \quad (4)$$

Applying the axisymmetry assumption, the volume element becomes $dV_0 = 2\pi R dR dZ$ and the surface element on the inner face, for $R = R_i$, equals $dA_0 = 2\pi R_i dZ$. Simplifying the common factor 2π , integration is reduced to the two-dimensional domain of the meridional plane Ω_{2D} .

Since the external load consists of a follower-type internal pressure, the reference stress vector is defined as $\mathbf{t}_{ref} = p_{in} J \mathbf{A}^{-T} \mathbf{N}$, where $\mathbf{N} = [-1, 0, 0]^T$ is the outward normal vector to the surface.

The final weak form, formulated for subsequent discretization and implementation in a nonlinear finite element solver, is expressed as follows:

$$\begin{aligned} \mathcal{R}(\mathbf{u}, p; \delta \mathbf{u}, q) = & \int_{\Omega_{2D}} (\mathbf{P}_{dev} \otimes \nabla_\chi \delta \mathbf{u}) R dR dZ \\ & - \int_{\Omega_{2D}} (p J \mathbf{A}^{-T} \otimes \nabla_\chi \delta \mathbf{u}) R dR dZ \\ & + \int_{\Omega_{2D}} (J - 1) q R dR dZ \\ & - \int_{\Gamma_{in}} (p_{in} J \mathbf{A}^{-T} \mathbf{N} \cdot \delta \mathbf{u}) R_i dZ = 0 \end{aligned} \quad (5)$$

Physical Meaning of Weak Form Terms

To clarify the formulation, the weak form can be decomposed into its various terms:

- **Deviatoric internal virtual work** ($\int \mathbf{P}_{dev} \otimes \nabla_\chi \delta \mathbf{u} R dR dZ$): Represents the elastic energy stored by the structure due to distortion, governed by the stiffness matrix of the Gent hyperelastic model.
- **Virtual work associated with hydrostatic pressure** ($\int -p J \mathbf{A}^{-T} \otimes \nabla_\chi \delta \mathbf{u} R dR dZ$): Constitutes the internal reaction necessary to maintain incompressibility, applying a hydrostatic stress p that resists volume changes.

- **Incompressibility equation in weak form** ($\int (J - 1)q R dR dZ$): Enforces that the determinant of the deformation gradient J equals 1 throughout the domain. This constraint is weakly satisfied by weighting it with the test function q , closing the mixed displacement-pressure system.
- **External virtual work for follower load** ($\int p_{in} J \mathbf{A}^{-T} \mathbf{N} \cdot \delta \mathbf{u} R_i dZ$): Identifies the work done by the pressure applied to the inner face. The Nanson term $J \mathbf{A}^{-T}$ maps the deformed normal to the reference configuration (pull back), ensuring that pressure always acts perpendicular to the current surface.

It is worth noting that the incompressibility constraint requires the problem to use a mixed u - p formulation so that the constraint is satisfied. However, this constraint is a weak constraint as it is not imposed as a condition in the problem but as a solution to the problem; therefore, in case of large geometric deformations, it is not guaranteed to be completely satisfied. It is thus important to monitor the value of J in the found solution to verify this constraint.

Results Analysis

Comparing FEM results with the analytical method, one can see that the two solutions coincide perfectly, thus verifying the accuracy of the weak formulation and finite element discretization. Looking more carefully at how the error varies with the deformed radius, an oscillatory behavior can be observed due to result interpolation between nodes; indeed, at nodes the error is greatly reduced compared to values along the cell. It is also interesting to note that the greatest error occurs for results in the tangential direction compared to results in the axial and radial directions. This is because there is greater nonlinearity in the tangential direction which, combined with interpolation issues, causes the interpolated value to deviate from the actual value.

